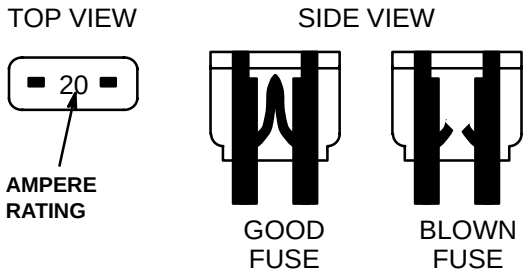
11-1 FUSE PANEL/CIRCUIT PROTECTION

CIRCUIT PROTECTION DEVICES

Electrical circuits on this vehicle may be protected by fuses, fusible links, fusible link cartridges, circuit breakers, or a combination of these devices.

BLADE TYPE FUSE

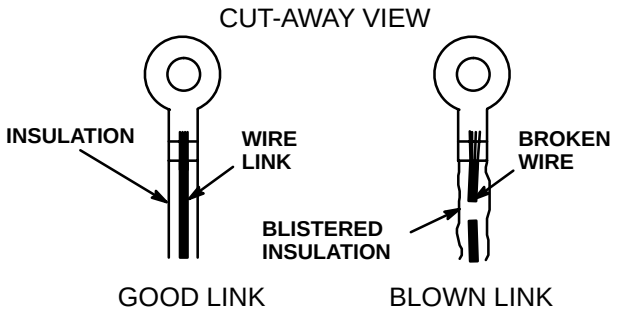


Blade type fuses have a transparent plastic housing. To check a fuse, pull it from the fuse panel and look at the fuse element through the housing. Always replace a blown fuse with a new fuse that has the same ampere rating.

The ampere rating of a blade type fuse can also be determined by following the color code shown here:

BLADE FUSE COLOR CODING	
AMPERE RATING	**HOUSING COLOR**
4	Pink
5	Tan
10	Red
15	Light Blue
20	Yellow
25	Natural
30	Light Green

FUSIBLE LINK



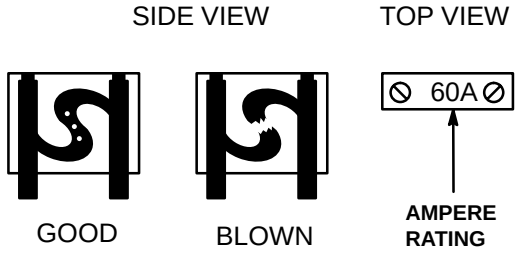
Fusible links are short lengths of wire that are smaller in diameter than the wires they are protecting. Fusible link wire is covered with a special thick, non-flammable insulation. An overload condition causes the insulation to blister. If the overload condition continues, the wire link will melt. To check a fusible link, look for blistered insulation. If the insulation is okay, pull lightly on the wire; If the fusible link stretches, the wire has melted.

When replacing fusible links, first cut the protected wire where it is connected to the fusible link. Then, tightly crimp or solder the new link to the protected wire.

Fusible links are often identified by color coding of the insulation, as shown here:

FUSIBLE LINK COLOR CODING	
WIRE LINK SIZE	**INSULATION COLOR**
20 GA	Blue
18 GA	Brown or Red
16 GA	Black or Orange
14 GA	Green
12 GA	Gray

MAXI-FUSE CARTRIDGE



Maxi–fuse cartridges have a transparent colored plastic housing. To check a maxi-fuse cartridge, look at the fuse element through the side of the housing.

To replace a maxi-fuse cartridge, pull it from the fuse box or panel. Always replace a blown maxi-fuse cartridge with a new one having the same ampere rating.

The ampere rating of a maxi-fuse cartridge can also be determined by following the color code shown here:

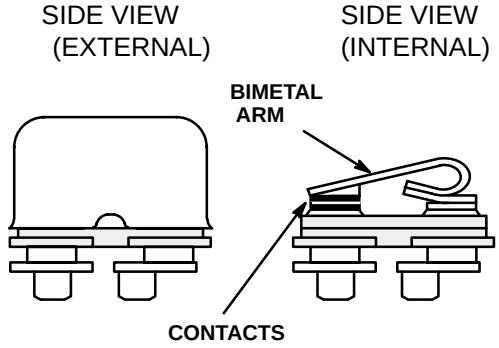
MAXI-FUSE CARTRIDGE COLOR CODING	
AMPERE RATING	**HOUSING COLOR**
30	Green
40	Orange
50	Red
60	Blue

CIRCUIT BREAKER

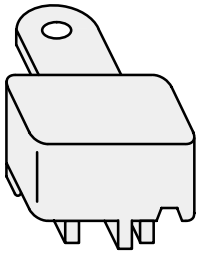
Some circuits are protected by circuit breakers (abbreviated “c. b.” in fuse chart).
They can be Fuse Panel mounted or in-line. Like fuses, they are rated in amperes.

Each circuit breaker conducts current through an arm made of two types of metal bonded together (bimetal arm). If the arm starts to carry too much current, it heats up. As one metal expands faster than the other the arm bends, opening the contacts. Current flow is broken. A circuit breaker can be the cycling or non-cycling type.

FUSE PANEL MOUNTED CYCLING TYPE

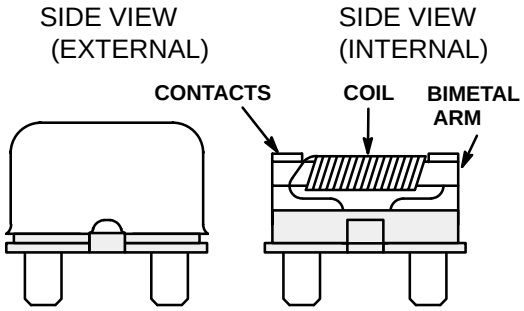


IN-LINE MOUNTED CYCLING TYPE

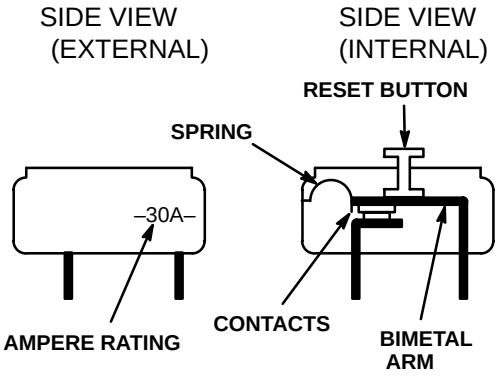


In the cycling type, the bimetal arm cools and straightens out. This cycle repeats as long as the overcurrent exists and power is applied.

FUSE PANEL MOUNTED NON-CYCLING TYPE



FUSE PANEL MOUNTED MANUAL RESET TYPE

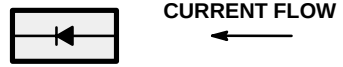


Two types of non-cycling circuit breakers are used; one is reset by removing power from the circuit, and the other is reset by depressing a reset button.

In the first type, there is a coil wrapped around the bimetal arm. When an overcurrent exists and the contacts open, a small current passes through the coil. This current through the coil is not enough to operate a load, but it does heat up both the coil and the bimetal arm. This keeps the arm in the open position until power is removed.

In the second type, a spring pushes the bimetal arm down and holds the contacts together. When an overcurrent condition exists and the bimetal arm heats up, the bimetal arm bends enough to overcome the spring and the contacts snap open. The contacts stay open until the reset button is pushed and the contacts snap together again.

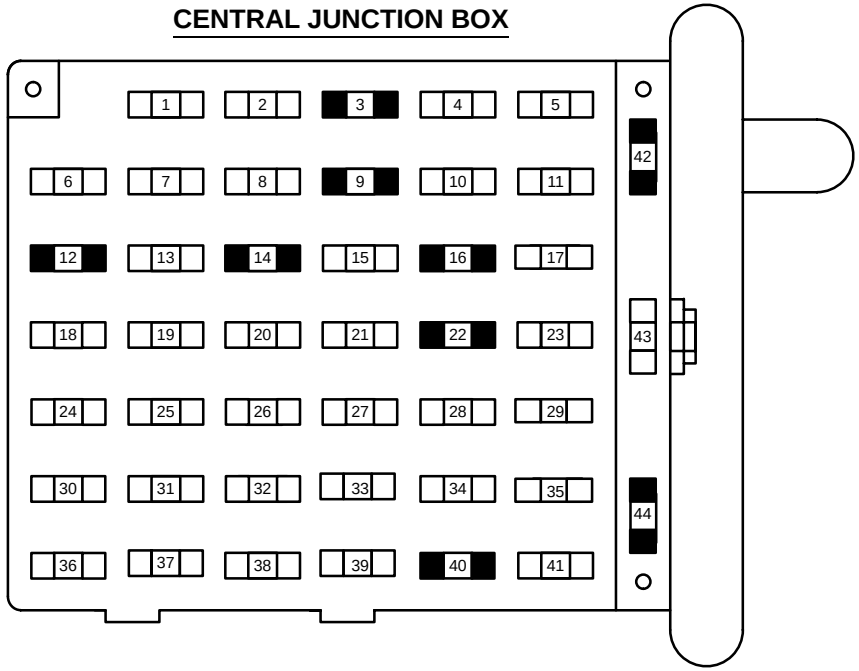
DIODE



Diodes are electrical devices that permit current to flow in one direction only. The current flows in the direction indicated by the arrow.

11-3 FUSE PANEL/CIRCUIT PROTECTION

CENTRAL JUNCTION BOX



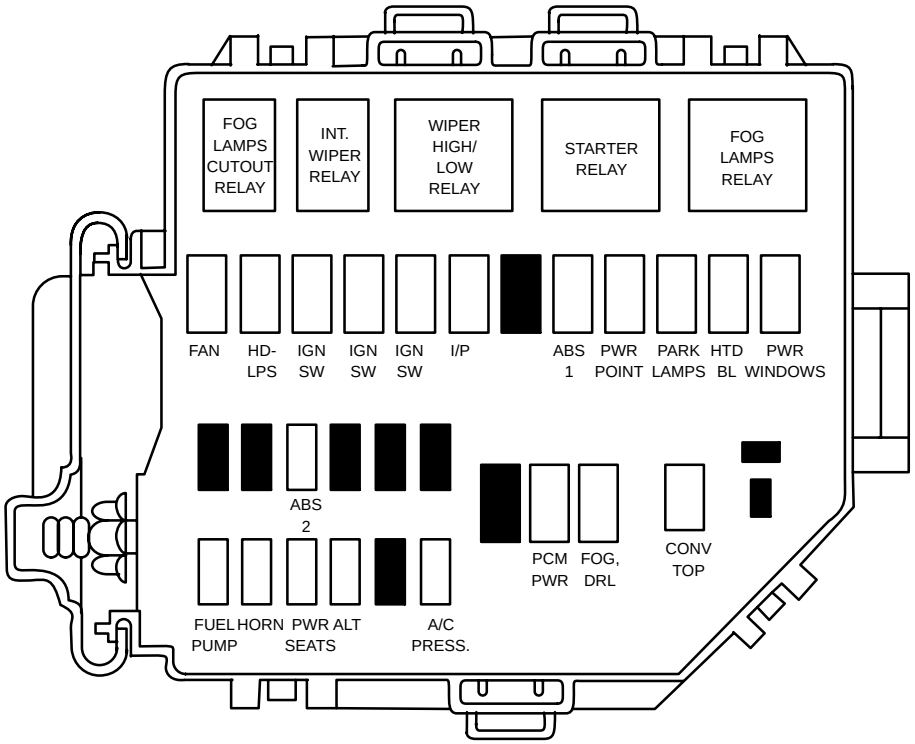
Fuse Value Amps	Color Code
4	Pink
5	Tan
10	Red
15	Light Blue
20	Yellow
25	Natural
30	Light Green

Fuse Position	Amps	Circuits Protected
1	20	Cigar Lighter
2	20	Engine Controls
3	–	NOT USED
4	10	RH Low Beam Headlamp
5	15	Instrument Cluster, Traction Control Switch
6	20	Starter Motor Relay
7	15	GEM, Interior Lamps
8	20	Engine Controls
9	–	NOT USED
10	10	LH Low Beam Headlamp
11	15	Reversing Lamps
12	–	NOT USED
13	15	Electronic Flasher
14	–	NOT USED
15	15	Power Lumbar Seats
16	–	NOT USED
17	15	Speed Control Servo, Shift Lock Actuator
18	15	Electronic Flasher
19	15	Exterior Rear View Mirror Switch, Power Locks, Park Lamp Relay
20	15	Convertible Top Switch
21	5	PCM, Instrument Cluster
22	–	NOT USED
23	15	A/C–Heater Control Assembly, Rear Window Defrost Control Switch

Fuse Position	Amps	Circuits Protected
24	30	A/C Heater Control Assembly
25	25	Luggage Compartment Lid Release Switch, Luggage Compartment Lid Relay
26	30	Wiper Motor, Wiper Relays
27	25	Radio
28	15	GEM, Transmission Control Switch
29	15	ABS Module
30	15	DRL Module
31	10	Data Link Connector
32	15	Radio, GEM
33	15	Brake Pressure Switch, Brake Pedal Position (BPP) Switch
34	20	Instrument Cluster, CCRM, PATS Transceiver Module, Ignition Coil, Right Radio Interface Capacitor
35	15	Shift Lock Actuator, PCM, Speed Control Servo, ABS Module
36	15	Restraints Control Module
37	5	Radio, A/C-Heater Control Illumination, Rear Window Defrost Control Switch, Traction Control Switch, Instrument Cluster, Fog Lamp Switch
38	20	High Beam Headlamps, Instrument Cluster, Fog Lamp Cutout Relay
39	5	GEM
40	–	NOT USED
41	15	Multifunction Switch/Stoplamps, High Mounted Stop/ Luggage Compartment
42	–	NOT USED
43	20 (c.b.)	GEM, Master Window/Door Lock Control Switch, Right Window/Door Lock Control Switch
44	–	NOT USED

11-5 FUSE PANEL/CIRCUIT PROTECTION

BATTERY JUNCTION BOX



Fuse Position	Amps	Circuits Protected
FAN (3.8L)	30 (c.b.)	Constant Control Relay Module (CCRM)
FAN (4.6L)	50 (Maxi)	Constant Control Relay Module (CCRM)
HD-LPS	30 (Maxi)	Main Light Swtich, Multifunction Switch
IGN SW	40 (Maxi)	Starter Motor Relay, Constant Control Relay Module (CCRM), GEM, Transmission Control Switch, Instrument Cluster
IGN SW	40 (Maxi)	ABS Module, Instrument Cluster, Reversing Lamps, Speed Control Servo, GEM, Shift Lock Actuator, A/C-Heater, Power Windows, Radio, Windshield Wiper Motor, Intermittent Wiper Relay, Wiper HIGH/LOW Relay
IGN SW	40 (Maxi)	Restraints Control Module, DRL Module, A/C-Heater, Blower Motor, Electronic Flasher
I/P	40 (Maxi)	Instrument Cluster, PCM
ABS1	50 (Maxi)	ABS Module
POWER POINT	20 (Maxi)	Auxiliary Power Point
PARK LAMPS	30 (Maxi)	Main Light Switch
HTD BL	30 (Maxi)	Rear Window Defrost Control Switch
POWER WINDOWS	40 (Maxi)	GEM
ABS2	20 (Mini)	ABS Module
FUEL PUMP	20 (Mini)	Constant Control Relay Module (CCRM)
HORN	20 (Mini)	Horns, Horn Relay
POWER SEATS	30 (Mini)	Driver Seat Control Switch
ALT	20 (Mini)	Generator
A/C PRESS.	20 (Mini)	Constant Control Relay Module (CCRM)
PCM POWER	30 (Maxi)	Constant Control Relay Module (CCRM)
CONV TOP	25 (c.b.)	Convertible Top Raise/Lower Relay
FOG, DRL	20 Maxi	Daytime Running Lamps (DRL) Module/Fog Lamps Relay